

Business Analytics (MBR2070T)

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Section: 2
Assigned by: SVC

Course: MBA
Roll No.: 71
Due Date: 02/03/2025

Identification of the Objectives:

Promoter's Perspective:

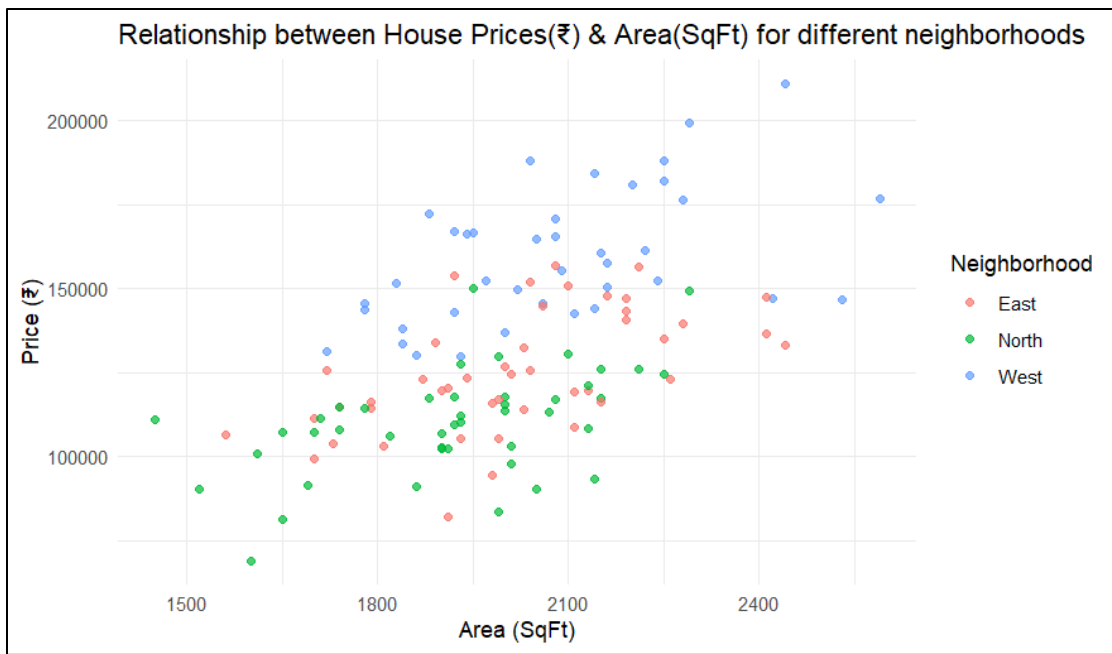
- Maximizing Profit: Determine which house features (e.g., sqFt, bedrooms, bathrooms, brick exterior, or neighborhood) correlate with higher sale prices to focus marketing efforts or set pricing strategies.
- Understanding Market Trends: Assess how the number of offers relates to price, as this could indicate demand and competition levels affecting sale outcomes.
- Identifying Sellable Features: Identify features associated with higher demand (e.g., via offers or price premiums) to prioritize in listings or developments.

Buyer's Perspective:

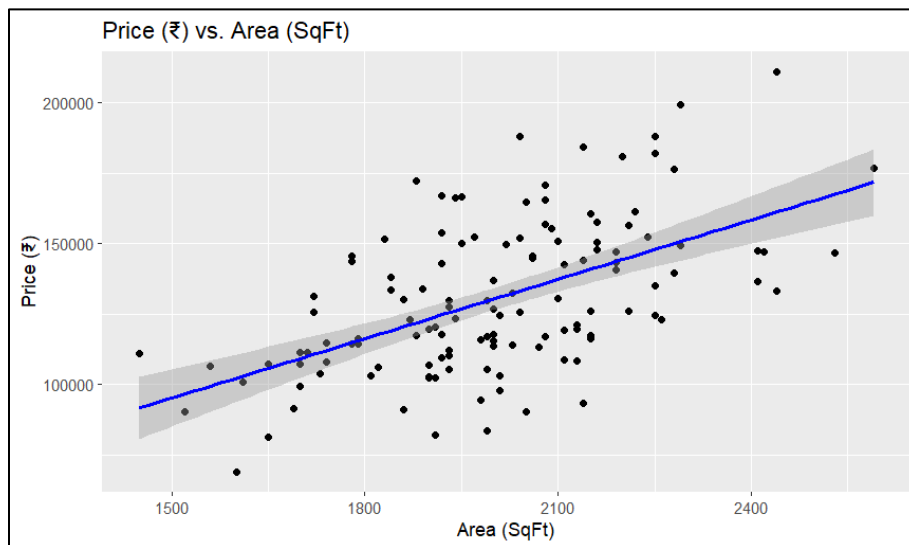
- Finding Best Value: Identify houses or neighborhoods offering the most area with respect to the money.
- Understanding Feature Costs: Quantify how much additional bedrooms, bathrooms, or a brick exterior increase the price to evaluate trade-offs.
- Assessing Market Competition: Examine how the number of offers influences price to gauge competition and negotiate effectively.

1. Scatter Plot of Price vs. Area (SqFt.) with Regression Line

```
1 attach(house_prices)
2 library(ggplot2)
3 df=data.frame(house_prices)
4 ggplot(df, aes(x = SqFt, y = Price, color = Neighborhood)) +
5   geom_point(alpha = 0.7)+
6   labs(title = "Relationship between House Prices(₹) & Area(SqFt) for different neighborhoods",
7         x = "Area (SqFt)",
8         y = "Price (₹)") + theme_minimal()
9
10 ggplot(df, aes(x = SqFt, y = Price)) +
11   geom_point() +
12   geom_smooth(method = "lm", color = "blue") +
13   labs(title = "Price (₹) vs. Area (SqFt)",
14         x = "Area (SqFt)",
15         y = "Price (₹)")
```



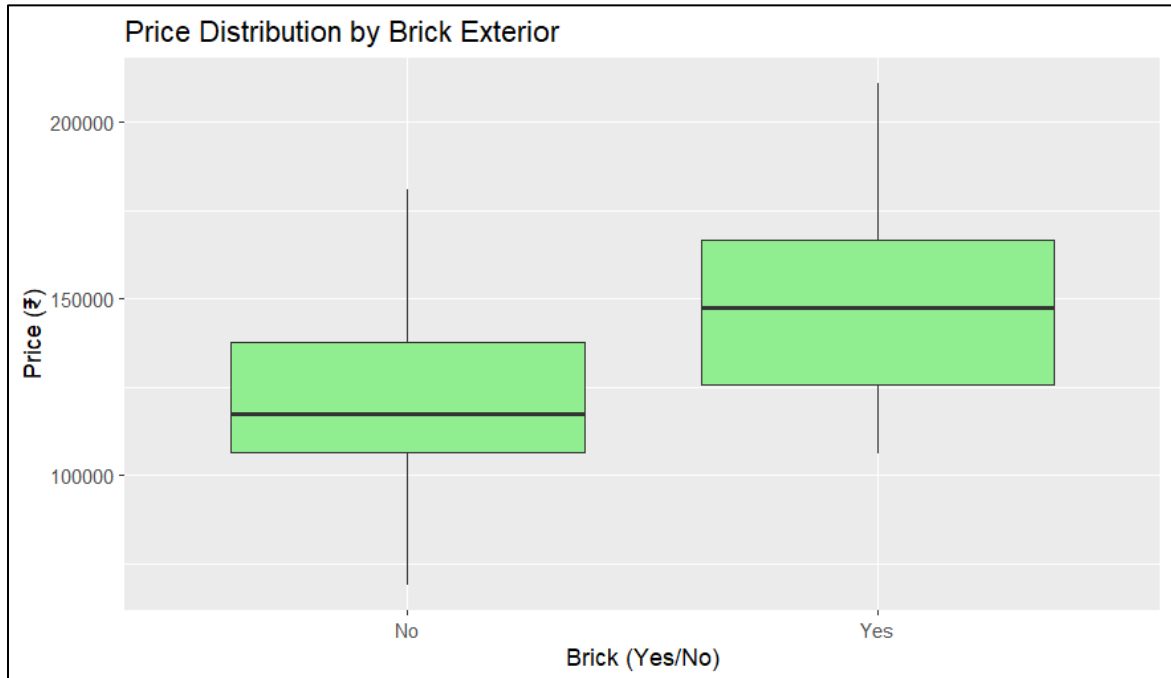
- There's a general positive correlation between house price and area (square footage).
- Larger homes tend to have higher prices.
- West (Blue): The blue points are generally clustered toward the higher price range, especially for larger areas. This suggests that homes in the West neighborhood are typically more expensive, possibly indicating a premium or wealthier area.
- East (Orange): The orange points are spread across a moderate price range (e.g., ₹100,000–₹160,000) and cover a wide range of areas. They appear to have a mix of lower and higher-priced homes, but fewer reach the highest prices compared to West.
- North (Green): The green points are mostly concentrated in the lower to mid-price range and smaller to medium areas (1,500–2,200 SqFt). This indicates that North is likely the most affordable neighborhood, with fewer expensive or large homes.



This graph shows a positive trend, with price increasing as area grows. The regression line indicates the average price increase per square foot. For promoters, this highlights the value of larger homes; for buyers, houses below the line may represent better value.

2. Box Plot of Price by Brick Exterior

```
16  
17 ggplot(df, aes(x = Brick, y = Price)) +  
18   geom_boxplot(fill = "lightgreen") +  
19   labs(title = "Price Distribution by Brick Exterior",  
20         x = "Brick (Yes/No)",  
21         y = "Price (₹)")
```



Houses with brick ("Yes") may have a higher median price. Promoters can emphasize brick exteriors; buyers can assess if the extra cost aligns with their preferences.

3. Bar Plot of Price by Number of Offers

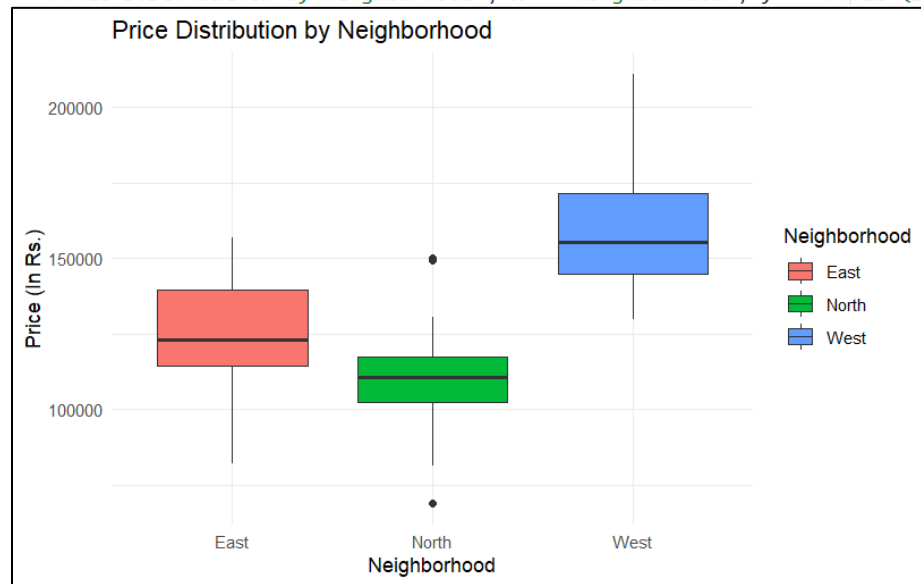
```
23 ggplot(df, aes(x = factor(Offers), y = Price)) +  
24   geom_boxplot(fill = "purple") +  
25   labs(title = "Price Distribution by Number of Offers",  
26         x = "Number of Offers",  
27         y = "Price (₹)")
```



Median prices might decrease slightly with more offers, suggesting competition drives prices down for affordable homes. Promoters can gauge demand; buyers assess competition intensity.

4. Boxplot of Neighborhood vs Price

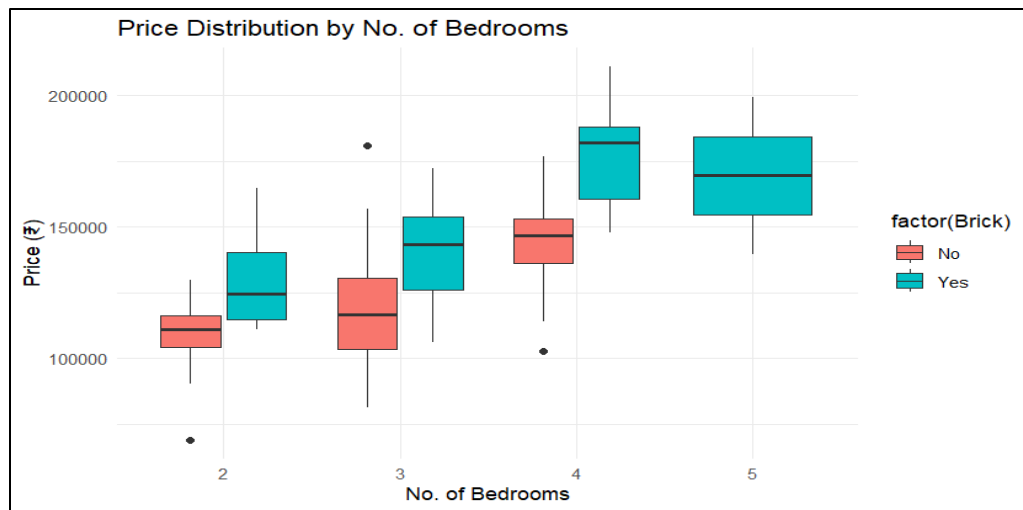
```
28  
29 ggplot(df, aes(x = Neighborhood, y = Price, fill = Neighborhood)) +  
30   geom_boxplot() + theme_minimal() +  
31   labs(title = "Price Distribution by Neighborhood", x = "Neighborhood", y = "Price (In Rs.)")
```



- West (Blue): The median price (the line inside the blue box) is the highest, approximately ₹160,000–₹170,000. This indicates that homes in the West neighborhood are, on average, the most expensive.
- East (Red): The median price for East is lower, around ₹120,000–₹130,000, suggesting it's a mid-range neighborhood in terms of price.
- North (Green): The median price for North is the lowest, approximately ₹100,000–₹110,000, indicating it's the most affordable neighborhood.
- Prices in West are skewed toward higher values, with the median and upper whisker reaching up to ₹200,000, indicating a premium market.
- East and North have lower median prices and narrower ranges, with North being the most affordable and least variable.
- The outlier in North suggests some variability or premium properties even in an otherwise affordable area.

5. Boxplot of No. of Bedrooms vs Price

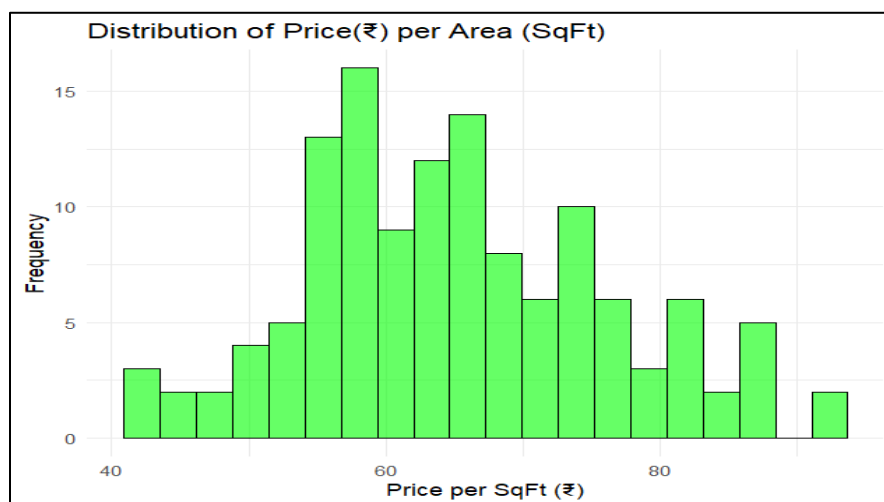
```
32  
33 ggplot(house_prices, aes(x = factor(Bedrooms), y = Price, fill = factor(Brick))) + geom_boxplot() +  
34   theme_minimal() +  
35   labs(title = "Price Distribution by No. of Bedrooms", x = "No. of Bedrooms", y = "Price (₹)")
```



- Across all bedroom counts, homes with a brick exterior (blue) consistently have higher median prices than non-brick homes (red). The difference is most pronounced for 3 and 4 bedrooms
- For 2 bedrooms, the difference is smaller, and the boxes overlap significantly, suggesting less of a premium for brick in smaller homes.
- For 5 bedrooms, the difference narrows, possibly due to fewer data points.
- Prices increase with the number of bedrooms, as expected in real estate markets. The jump from 4 to 5 bedrooms is particularly visible, suggesting a premium for larger homes.
- Brick exteriors consistently add value, but the premium diminishes slightly for 5-bedroom homes, possibly due to other factors (e.g., location, area) overshadowing brick status.

6. Histogram of Average Price(₹) per Area (SqFt)

```
51 house_prices$Price_per_AreaSqFt <- (house_prices$Price / house_prices$SqFt)
52 ggplot(df, aes(x = Price_per_AreaSqFt)) + geom_histogram(bins = 20, fill = "green", color = "black", alpha = 0.6)
53 + theme_minimal()
54 + labs(title = "Distribution of Price(₹) per Area (SqFt)", x = "Price per SqFt (₹)", y = "Frequency")
```



- The histogram is right-skewed (positively skewed), meaning most houses have a lower price per SqFt., with fewer houses having higher prices per SqFt.
- The highest frequency bar is around ₹50–₹60/SqFt, with a frequency of approximately 14–15 houses. This suggests that the majority of homes in the dataset fall within this price range per SqFt.
- Beyond ₹70/SqFt, the frequency drops sharply, with only 1–2 houses at ₹70–₹80/SqFt, indicating that very few homes have a high price per SqFt..
- The mode appears to be around ₹50–₹60/SqFt, suggesting this is the typical price per square foot for most houses.